



# SIMATS Online Programming Lab

# JAVA PROGRAMMING

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## Q2

Hospital Token Management  
System using List Interface  
10 marks

**Q3** 

Airport Boarding Queue  
Management using Queue  
Interface

10 marks

#### Q4

Digital Certificate Verification  
using Set Interface  
10 marks

**Q5** 

Smart Inventory Management  
using Map Interface

10 marks

5/5 answered

3. Replace every occurrence of a given word with another word.
4. Reverse each sentence individually.
5. Maintain editing history using `StringBuffer`.
6. Construct the final formatted document using `StringBuilder`.
7. Display the original document, editing history, and final document.

Requirement: Use `String` methods wherever appropriate. Use `StringBuilder` for document construction and `StringBuffer` for edit history.

## Your Code

### Input

Java is a very powerful language and it can be used to do anything you want including creating web application and databases

## Output





10 marks

**Q5** 

Smart Inventory Management  
using Map Interface

10 marks

## Hospital Token Management System using List Interface

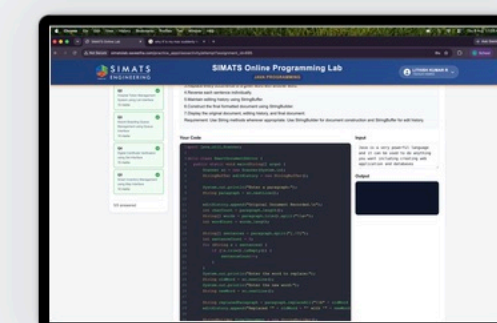
- Store patient details using an appropriate List implementation.
- Register new patients.
- Display patients in registration order.
- Update patient information using Patient ID.
- Remove cancelled appointments.
- Display all patients of a particular department.
- Find the first and last registered patients.

Requirement: Use a suitable List implementation

### Input

## Output

\_\_\_\_\_





10 marks

### Q5

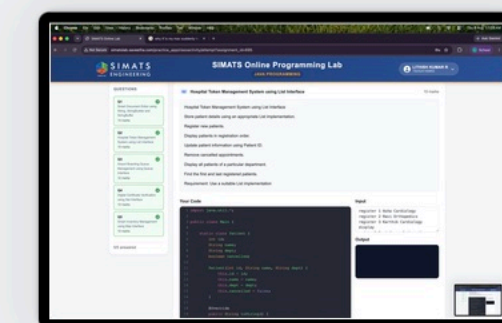
Smart Inventory Management  
using Map Interface  
10 marks

Requirement: Use appropriate Queue implementations.

### Input

```
3
Ravi
Yes
5
```

## Output





Remove revoked certificates.

Requirement: Use suitable Set implementations.

```

1 import java.util.*;
2 public class DigitalCertificateVerification {
3     private static final Set<String> insertionOrderSet = new LinkedHas
4     private static final Set<String> sortedOrderSet = new TreeSet<>();
5     private static final Set<String> revokedSet = new HashSet<>();
6
7     public static void addCertificate(String certId) {
8         if (certId == null || certId.trim().isEmpty()) return;
9
10        certId = certId.trim();
11
12        boolean addedInsertion = insertionOrderSet.add(certId);
13        sortedOrderSet.add(certId);
14
15        if (addedInsertion) {
16            System.out.println("Certificate added: " + certId);
17        } else {
18            System.out.println("Duplicate entry ignored: " + certId);
19        }
20    }
21
22    public static void verifyCertificate(String certId) {
23        if (certId == null || certId.trim().isEmpty()) return;
24
25        certId = certId.trim();
26
27        if (revokedSet.contains(certId)) {
28            System.out.println("Certificate REVOKED: " + certId);
29            return;
30        }
31
32        if (insertionOrderSet.contains(certId)) {
33            System.out.println("Certificate VERIFIED: " + certId);
34        } else {
35            System.out.println("Certificate NOT FOUND: " + certId);

```

```
1
C1
1
C2
2
```

```
===== Digital Certificate
Verification (Set) =====
1. Add certificate ID
2. Verify certificate ID
3. Display certificates (insertion)
```



← Back

### Q5 Smart Inventory Management using Map Interface

10 marks

## Q2

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10 marks

#### Q4

Digital Certificate Verification  
using Set Interface  
10 marks

**Q5** 

Smart Inventory Management  
using Map Interface

10 marks

5/5 answered

## Smart Inventory Management using Map Interface

- Store product information using an appropriate Map implementation.
- Add new products.
- Update product quantity.
- Search a product using Product ID.
- Calculate the total inventory value.
- Display products category-wise.
- Remove discontinued products.
- Display all available products.

Requirement: Use a suitable Map implementation.

## Your Code

```
1 import java.util.*;
2
3 public class Main {
4     Map<String, Product> inv = new HashMap<>();
5
6     void add(Product p) {
7         inv.put(p.id, p);
8     }
9
10    void updateQty(String id, int q) {
11        if (inv.containsKey(id)) inv.get(id).qty = q;
12    }
13
14    Product find(String id) {
15        return inv.get(id);
16    }
17 }
```

### Input

```
stdin input
```

## Output

```
Search P103: ID=P103, Name=Cheese,  
Category=Dairy, Price=180.00,  
Qty=8, Discontinued=false  
Total Value: 3590.0  
  
--- Category Wise Available
```